

Causal Inference: Modern Methods

Problem Set

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Abstract

This problem set covers modern methods in causal inference, including potential outcomes, directed acyclic graphs (DAGs), instrumental variables, regression discontinuity, and difference-in-differences. Students will gain practical experience identifying causal effects from observational data and understanding threats to causal identification.

1 Potential Outcomes Framework

Exercise 1 (Fundamental Problem of Causal Inference). *Consider a job training program where $D_i = 1$ if individual i receives training and $D_i = 0$ otherwise. Let $Y_i(1)$ be the potential outcome with training and $Y_i(0)$ without.*

- (a) *Define the individual treatment effect (ITE), average treatment effect (ATE), and average treatment effect on the treated (ATT).*
- (b) *Explain why we cannot observe $Y_i(1) - Y_i(0)$ for any individual.*
- (c) *Show that the naive comparison $E[Y_i|D_i = 1] - E[Y_i|D_i = 0]$ equals:*

$$\underbrace{E[Y_i(1) - Y_i(0)|D_i = 1]}_{ATT} + \underbrace{E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0]}_{\text{Selection Bias}}$$

- (d) *Under what conditions does the naive comparison identify the ATE?*

Exercise 2 (Conditional Independence Assumption). *The conditional independence assumption (CIA) or unconfoundedness states:*

$$\{Y_i(0), Y_i(1)\} \perp D_i | X_i$$

- (a) *Explain this assumption in plain language.*
- (b) *Why is it called "unconfoundedness"?*
- (c) *Give an example where CIA is plausible.*
- (d) *Give an example where CIA is clearly violated.*
- (e) *Under CIA, show that:*

$$E[Y_i(1) - Y_i(0)|X_i] = E[Y_i|D_i = 1, X_i] - E[Y_i|D_i = 0, X_i]$$

2 Directed Acyclic Graphs (DAGs)

Exercise 3 (Drawing and Interpreting DAGs). *For each scenario, draw a DAG and identify the causal effect of interest:*

(a) **Education and Earnings:**

- *Treatment: Years of education (D)*
- *Outcome: Annual earnings (Y)*
- *Confounders: Ability (A), Family background (F)*
- *Mediator: Job type (J)*

Draw the DAG and identify:

- *All paths from D to Y*
- *Which paths are backdoor paths?*
- *What variables should be controlled?*
- *What is the direct effect of D on Y (not through J)?*

(b) **Collider Bias:** *Consider $D \rightarrow Y \leftarrow U$ where U is unobserved.*

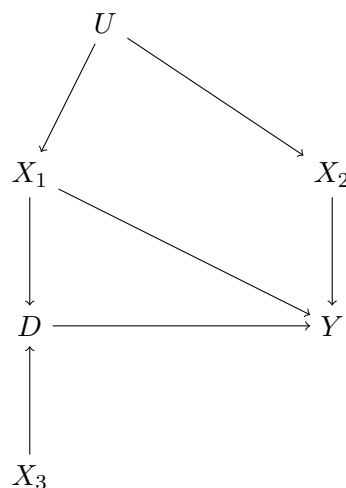
- *Explain why conditioning on Y induces association between D and U*
- *Provide a real-world example of collider bias*

(c) **M-Bias:** *Draw a DAG with structure: $U_1 \rightarrow D$, $U_2 \rightarrow Y$, $U_1 \rightarrow M \leftarrow U_2$.*

- *Is there an open backdoor path from D to Y ?*
- *What happens if you control for M ?*
- *When is controlling for M harmful?*

Problem 1 (Backdoor Criterion). *Use Pearl's backdoor criterion to determine the minimal sufficient adjustment set.*

Scenario: *Consider the following DAG:*



1. *List all paths from D to Y*
2. *Identify backdoor paths*
3. *Find all valid adjustment sets*
4. *What is the minimal sufficient adjustment set?*
5. *Can we identify the causal effect if U is unobserved?*

3 Instrumental Variables

Problem 2 (Returns to Education). *Estimate the returns to education using quarter of birth as an instrument.*

Background: Angrist & Krueger (1991) used quarter of birth as an instrument for education, exploiting compulsory schooling laws.

Dataset: Simulate or use provided data with:

- Y : Log annual earnings
- D : Years of education
- Z : Quarter of birth indicator (born in Q1)
- X : Controls (age, region, etc.)

Tasks:

1. **Motivation:**

- Why is education endogenous?
- Draw a DAG showing the endogeneity problem
- Explain how quarter of birth could be a valid instrument

2. **Validity Checks:**

- Test relevance: Regress D on Z (first stage)
- Compute first-stage F -statistic
- Discuss exclusion restriction: Why might it fail?

3. **Estimation:**

- Estimate OLS (biased)
- Estimate 2SLS
- Compare coefficients
- Compute heteroskedasticity-robust standard errors

4. **Interpretation:**

- What is the local average treatment effect (LATE)?
- Who are the compliers?
- Why might IV estimate differ from OLS?

5. **Sensitivity:**

- Add/remove controls
- Test overidentification (if multiple instruments)
- Discuss weak instruments problem

Problem 3 (Weak Instruments). *Investigate the consequences of weak instruments.*

Simulation:

1. Generate data: $n = 500$, instrument strength $\gamma \in \{0.05, 0.1, 0.3, 0.5\}$

$$D = \gamma Z + \alpha U + \epsilon_D$$

$$Y = \beta D + \alpha U + \epsilon_Y$$

where U is unobserved confounder.

2. For each γ :

- Estimate 2SLS
- Compute first-stage F -statistic
- Repeat 1000 times (Monte Carlo)

3. Analyze:

- Bias of 2SLS estimator
- Coverage of 95% confidence intervals
- How weak is "weak"? (F ; 10 rule of thumb)

4. Propose solutions:

- Limited information maximum likelihood (LIML)
- Anderson-Rubin test
- Weak instrument robust inference

4 Regression Discontinuity Design

Problem 4 (Class Size and Student Achievement). Use RDD to estimate the effect of class size on test scores.

Setup: Schools with enrollment above 40 students must split into two classes.

Data:

- Running variable: School enrollment (around cutoff of 40)
- Treatment: Small class (enrollment ≤ 40) vs. large class
- Outcome: Average test score

Tasks:

1. **Identification:**

- State the RDD identification assumptions
- Why is continuity of potential outcomes crucial?
- Draw a hypothetical RDD plot

2. **Validity Checks:**

- Density test: Check for manipulation of running variable (McCrary test)
- Covariate balance: Test continuity of pre-treatment covariates
- Placebo cutoffs: Test at other points

3. **Estimation:**

- *Local linear regression (both sides of cutoff)*
- *Optimal bandwidth selection (Imbens-Kalyanaraman or MSE-optimal)*
- *Robust inference (bias-corrected CI)*

4. **Sensitivity:**

- *Bandwidth sensitivity plot*
- *Polynomial order (linear vs. quadratic)*
- *Donut RDD (exclude observations near cutoff)*

5. **Interpretation:**

- *Is this effect local or global?*
- *What is the relevant population?*
- *External validity concerns*

Problem 5 (Fuzzy RDD). *Extend to fuzzy RDD where treatment uptake is imperfect.*

Scenario: *Assignment to small class at cutoff, but:*

- *Some schools just above cutoff have small classes (non-compliance)*
- *Some schools just below have large classes*

Tasks:

1. *Explain the difference between sharp and fuzzy RDD*
2. *Show that fuzzy RDD can be seen as an IV problem*
3. *Estimate both the "first stage" (effect on treatment) and "reduced form" (effect on outcome)*
4. *Compute the fuzzy RDD estimand (ratio)*
5. *Interpret as local average treatment effect (LATE)*

5 Difference-in-Differences

Problem 6 (Minimum Wage and Employment). *Replicate Card & Krueger (1994) analysis of minimum wage effects.*

Setup:

- *Treatment: New Jersey minimum wage increase (April 1992)*
- *Control: Pennsylvania (no increase)*
- *Outcome: Employment at fast-food restaurants*
- *Periods: Before (Feb 1992) and After (Nov 1992)*

Tasks:

1. **Identification:**
 - *State parallel trends assumption*
 - *Why is it crucial?*

- Draw hypothetical trend plots

2. **Estimation:**

- Compute 2x2 DiD estimator (manual calculation)
- Estimate regression specification:

$$Y_{it} = \alpha + \beta \text{Treat}_i + \gamma \text{Post}_t + \delta(\text{Treat}_i \times \text{Post}_t) + \epsilon_{it}$$

- Interpret δ (the DiD estimator)
- Compute cluster-robust standard errors (at state level)

3. **Parallel Trends Testing:**

- Test for pre-treatment differential trends
- Event study plot (if multiple time periods available)
- Placebo tests using pre-treatment periods

4. **Extensions:**

- Add covariates
- Triple differences (add another control group)
- Heterogeneous treatment effects by restaurant characteristics

Problem 7 (Staggered Adoption). Analyze staggered treatment adoption with recent methods.

Setup: Multiple states adopt minimum wage increases at different times.

Problem: Two-way fixed effects (TWFE) can give negative weights to some treatment effects (Goodman-Bacon decomposition).

Tasks:

1. Simulate staggered adoption data with heterogeneous effects
2. Estimate naive TWFE:

$$Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \epsilon_{it}$$

3. Show that TWFE can be biased with:

- Treatment effect heterogeneity
- Dynamic effects

4. Implement modern alternatives:

- Callaway & Sant'Anna (2021)
- Sun & Abraham (2021)
- Borusyak et al. (2021)

5. Compare all estimates and discuss

6 Matching and Propensity Scores

Problem 8 (Propensity Score Matching). *Estimate treatment effects using propensity score methods.*

Scenario: *Effect of college degree on earnings.*

Data:

- *Treatment: College degree (vs. high school)*
- *Outcome: Annual earnings*
- *Covariates: Age, gender, parents' education, test scores, etc.*

Tasks:

1. Propensity Score Estimation:

- *Fit logistic regression: $P(D = 1|X)$*
- *Check for common support*
- *Plot propensity score distributions by treatment group*

2. Matching Methods:

- *Nearest neighbor matching (1:1, 1:k)*
- *Caliper matching*
- *Kernel matching*

3. Balance Checking:

- *Standardized mean differences before/after matching*
- *Love plot*
- *Variance ratios*

4. Estimate ATT:

- *Using matched samples*
- *IPW (inverse probability weighting)*
- *Doubly robust estimation*

5. Sensitivity Analysis:

- *Rosenbaum bounds*
- *How strong would hidden bias need to be?*

7 Synthetic Control Methods

Problem 9 (Policy Evaluation). *Use synthetic control method to evaluate a policy intervention.*

Example: *German reunification effect on West German GDP (Abadie et al. 2015).*

Tasks:

1. *Explain the synthetic control idea*

2. *Construct synthetic control as weighted average of donor units:*

$$\hat{Y}_{0t} = \sum_{j=2}^{J+1} w_j Y_{jt}$$

where weights minimize pre-treatment fit

3. *Estimate treatment effect: $\hat{\tau}_t = Y_{1t} - \hat{Y}_{0t}$*
4. *Placebo tests: Apply method to control units*
5. *Inference: Permutation-based p-values*
6. *Sensitivity: Leave-one-out analysis*

8 Submission Guidelines

8.1 Format

- Python (Jupyter notebook) or R (R Markdown)
- Include all code, results, and visualizations
- Clear explanations and interpretations
- Well-documented code

8.2 Evaluation

- **Identification (30%):** Understanding of causal assumptions
- **Implementation (30%):** Correct estimation procedures
- **Interpretation (25%):** Thoughtful discussion of results
- **Validity Checks (15%):** Robustness and sensitivity analyses

9 Resources

Code: code/causal_inference/

Bibliography: shared/bibliographies/causal_inference_references.bib

Key Papers:

- Angrist & Pischke (2008). *Mostly Harmless Econometrics*
- Pearl (2009). *Causality*
- Imbens & Rubin (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences*